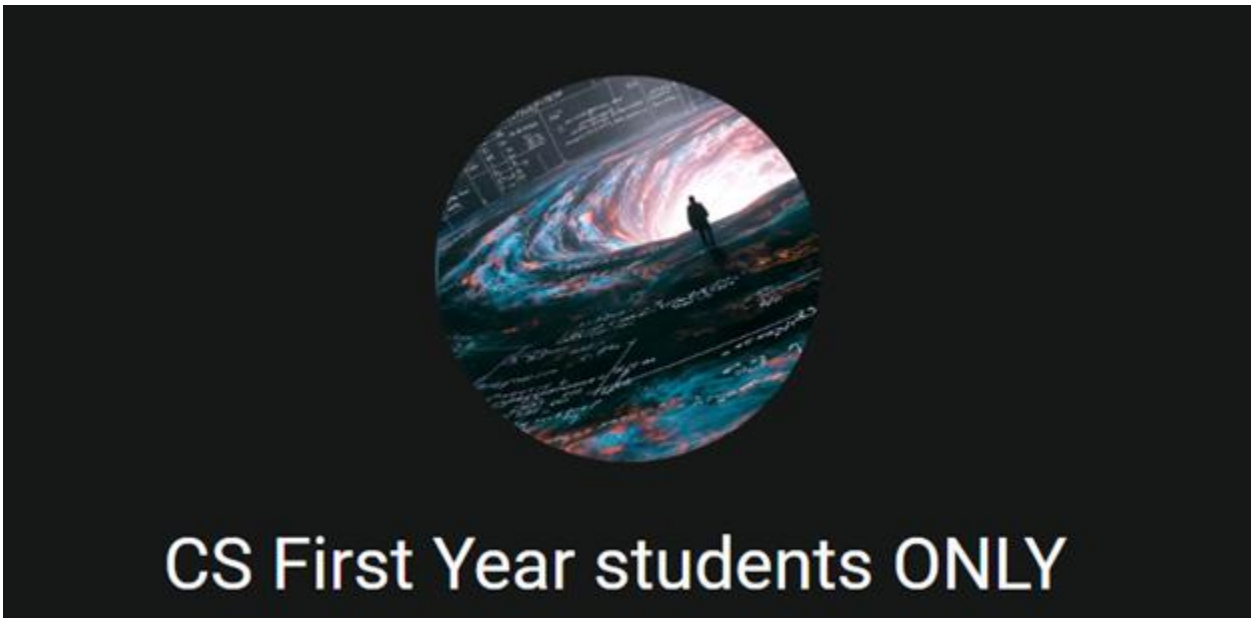
Midterm Exam For Second Semester (2024/2025)



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Midterm Exam for Second Semester (2024/2025)

Course Name: Object Oriented Programming	**Course Code: CS103**	**Bylaw: 2024**
Year: First	**Program: Computer Science**	
Date of Exam: Saturday 22 /3/2025	**Time Allowed: 1-hour**	**Total Marks: 20**
Course Instructor: Prof. Mohamed Sayed Farag	**No. of questions: (3)**	**No. of pages: (1)**

Exam instructions

- **Make sure that the question form is identical to the answer sheet form**
- **The answer sheet must be in good condition from the moment received until it is delivered after taking the exam**
- **Do not bend the answer sheet or lean on it while answering**

Answer the following questions:

Question One: [4 Marks], 0.5 for each. (ILOs: a1, a2, b1)

Choose the correct answer from (A), (B), (C) or (D):

1	Which feature of OOPS described the reusability of code?			
	A	Abstraction	**B**	Encapsulation
	C	Polymorphism	**D**	Inheritance
2	Which of the following is not an OOP?			
	A	Java	**B**	C#
	C	C++	**D**	C
3	OOP acronym for			
	A	Object of Programming	**B**	Object Original Programming
	C	Object Oriented Programming	**D**	Operating Original Programming
4	Which feature of OOPS derives the class from another class?			
	A	Inheritance	**B**	Data hiding
	C	Encapsulation	**D**	Polymorphism
5	Which of the following is correct about class?			
	A	class can have member functions while structure cannot.	**B**	class data members are public by default while that of structure are private.
	C	Pointer to structure or classes cannot be declare	**D**	class data members are private by default while that of structure are public by default
6	Which of the following gets called when an object is being created?			
	A	Constructor	**B**	Virtual Function
	C	Destructors	**D**	Main
7	Which among the following features do not come under the concept of OOPS?			
	A	Platform independent	**B**	Data binding
	C	Data hiding	**D**	Message passing
8	Which syntax among the following shows that a member is private in a class?			
	A	private :: Name(parameters)	**B**	private: function Name(parameters)
	C	private(function Name(parameters))	**D**	private function Name(parameters)

Question Two: [10 Marks], 2 for each (ILOs: a1, a2, b1, b2, c1)

Define the following concepts:

Class – Object – Encapsulation – Constructors – Destructors

Class: class represents a set of related objects; Classes are user defined data types that behave like the built-in types of programming language.

It is the blueprint for an object in object-oriented programming language.

Object: Objects are the basic run-time entities in an object-oriented system "it's an instance of a class".

Encapsulation: encapsulation is the concept of wrapping together of data and information in a single unit.

Constructors: Constructors are special class functions which perform initialization of every object. Constructor has the same name as the class. Constructors are public in the class. Constructor does not have any return type.

Destructors: Destructors are typically used to de-allocate memory. Also, they are used to clean up for objects and class members when the object gets terminated.

Question Three: [6 Marks], 3 for each (ILOs: a2, b1, b2, c1, c4)

What will be **the output** of the following programs?

A

```
#include <iostream>
using namespace std;
class Point {
   Point() { cout << "Constructor called";
            }
};
int main()
{
  Point t1;
  return 0;
}
```

Constructor called

B

```
#include <iostream>
using namespace std;
class demo {
      public:
          int f_num, s_num;
          void sum(int a, int b)
               { cout <<a+b;
           } };
int main(){
 demo d1;
d1.sum(d1.f_num=10, d1.s_num=20);
return 0;
}
```

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End of Questions
Good Luck,
Prof. Mohamed Sayed Farag